Practical Questions

- <u>Output questions, what is the out put of the following:</u>

```
class Student{
private:
    int id;
    string name;
public:
    Student (int id, string name) {
     this.id=id;
     this.name=name;
     cout<<this.id<<":"<<this.name;
     }
};
int main (){
 Student s1(1,"Ali").
}
```

Output

```
class Circle{
private:
    double radius;
public:
    void setRadius(double r){
     radius=r; }
    double getRadius()const{
     return radius; }
};
int main(){
 Circle c1;
 c1.radius=5.5;
 cout<<getRadius();
}
```

Output

```
class Testclass{
private:
    static int counter;
public:
    Testclass (){
        cout<<"Const \n";
        counter+=3; }
    ~ Testclass (){
        cout<<"Destr \n"; }
    void display() const{
        cout<<counter<<endl; }
};
int Testclass::counter=3;
int main()
{
 {
  Testclass t1;
  t1.display();
 }
 Testclass t2;
 t2.display();
}
```

Output

```
class Distance{
    int feet,inches;
public:
    Distance(int f,int i){
        feet=f, inches=i; }
    void Setdistance(int f,int i){
        cout<<"SetDis func \n";
        feet=f, inches=i; }
    void display() const {
        cout<<feet<<inches<<endl; }
};
int main()
{
   const Distance dis(5,10);
   dis.Setdistance(7,7);
   dis.display();
}
```

Output

Each of the following class declarations has errors. Locate as many as you can.

```
class Box{
private:
    double width;
    double length;
    double height;
public:
    Box(double w, l, h) { width = w; length = l; height = h; }
    Box(Box b) {   //copy constructor
        width = b.width;
       length = b.length;
       height = b.height; }
};
```

Find the error(s) in each of the following and explain how to correct it (them):

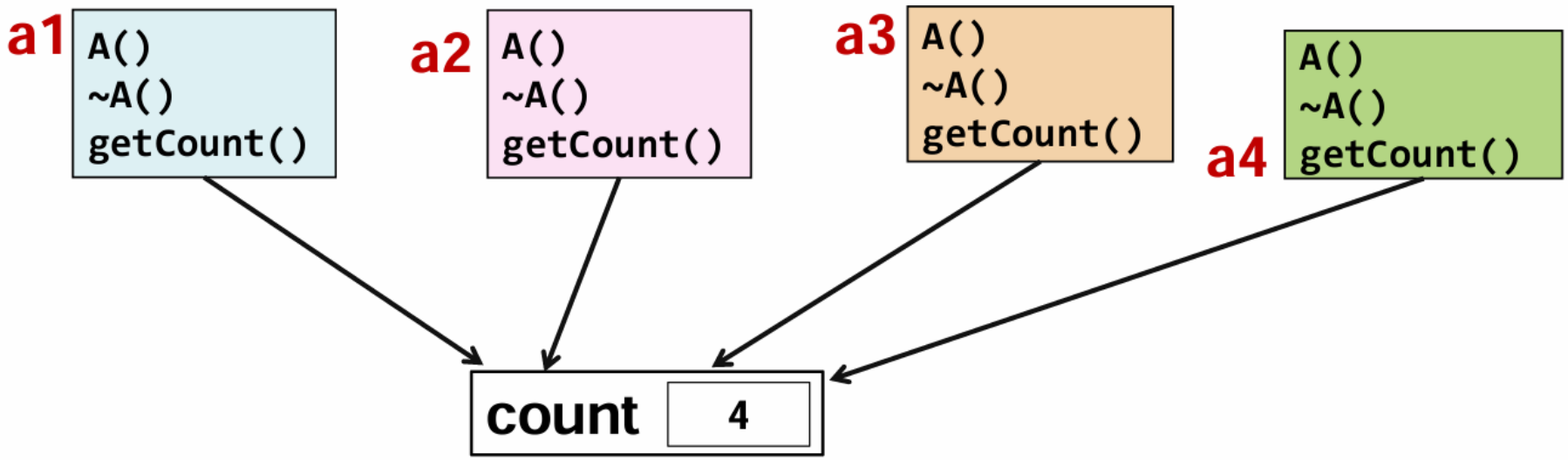
a) Assume the following prototype is declared in class `Time`:

```
void ~Time( int );
```

c) Assume the following prototype is declared in class `Employee`:

```
int Employee( string, string );
```

- The following draw indicate Class A with 4 different object, they all refer to the same attribute count, <u>write line of code for count definition inside class</u>.



What will the following program display on the screen?

```
#include <iostream>
using namespace std;
class Tank {
 private:
  int gallons;
 public:
  Tank() {
    gallons = 50;
  }
  Tank(int gal) {
    gallons = gal;
  }
  int getGallons() {
    return gallons;
  }
};
int main() {
  Tank storage[3] = { 10, 20 };
  for (int index = 0; index < 3; index++)
    cout << storage[index].getGallons() << endl;
}
```

- Write simple code for the following:

Define a dynamic array called **Darr** of 500 integers initialized to zero.

You have a **Distance** class that has **feet** as int, and **inches** as float, Based on this:

2. Define an overloaded constructor that sets **ft** parameter to feet and **inc** param to inches. (Initialization Must be in the same line using **initializer list** not in the body of constructor).

3. Define an overloaded constructor (with **default values** 1 and 1) that set **feet** parameter to feet attribute and **inches** param to inches attribute. (initialization Must be in the body of constructor).

4. Define an array of **3 Distance class objects**, initializing each one using the overloaded constructor with the values (2,3), (5,3), and (7,3) in the **initializer list**.

```
1  class Point{
2      private:
3          int x;
4          int y;
5      public:
6          Point(int x=0, int y=0): x(x),y(y){}
7
8          void setX(int x){this->x=x;}
9          int getX(){return x;}
10         void setY(int y){this->y=y;}
11         int getY(){return y;}
12 };
13
```

- Write array of Points with size 3 using list initializer, where first item has 5,5 – second item has 1,2 and third item is default
- Create pointer to object Point with default values.

- Which line number having overloaded constructor?
- Modify overloaded constructor to initialize attributes inside body using this
- Write code for copy constructor
- Modify the code to count number of created object from this class (attribute + method to get count).

…… is a member function “has the same name of class”, it hasn’t a return data type, as it can’t return anything, not even void.

A) Constructor B) Setter function C) Destructor D) Const function

…… data item doesn’t belong to a single instance of a class, and it is useful when all objects of the same class must share a common item of information.

A) Protected B) Constant C) Static D) Shared

In class definition, data or functions designated privately are accessible

A) to any function in program B) only if you know the password.

C) to member functions of that class D) only to public members of the class

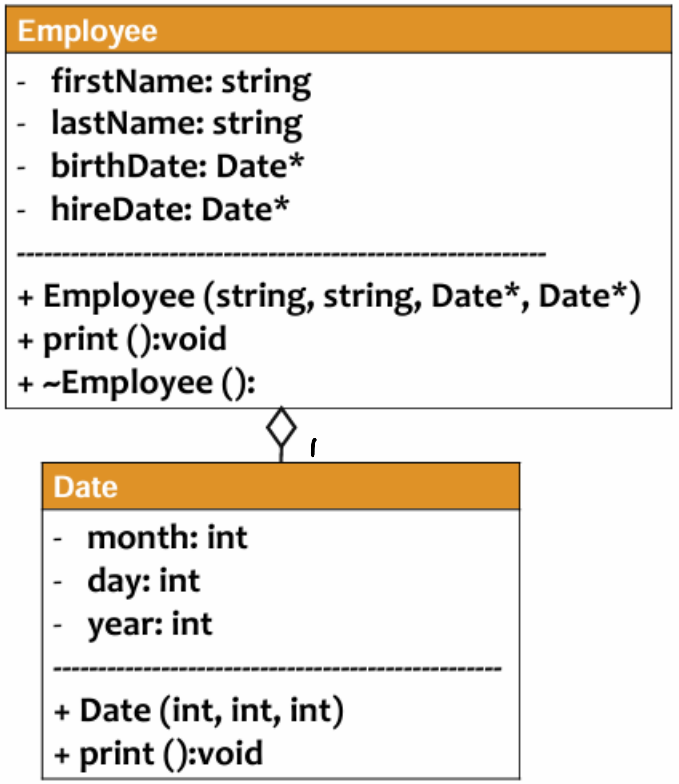
The dot operator connects the following two entities (reading from left to right).

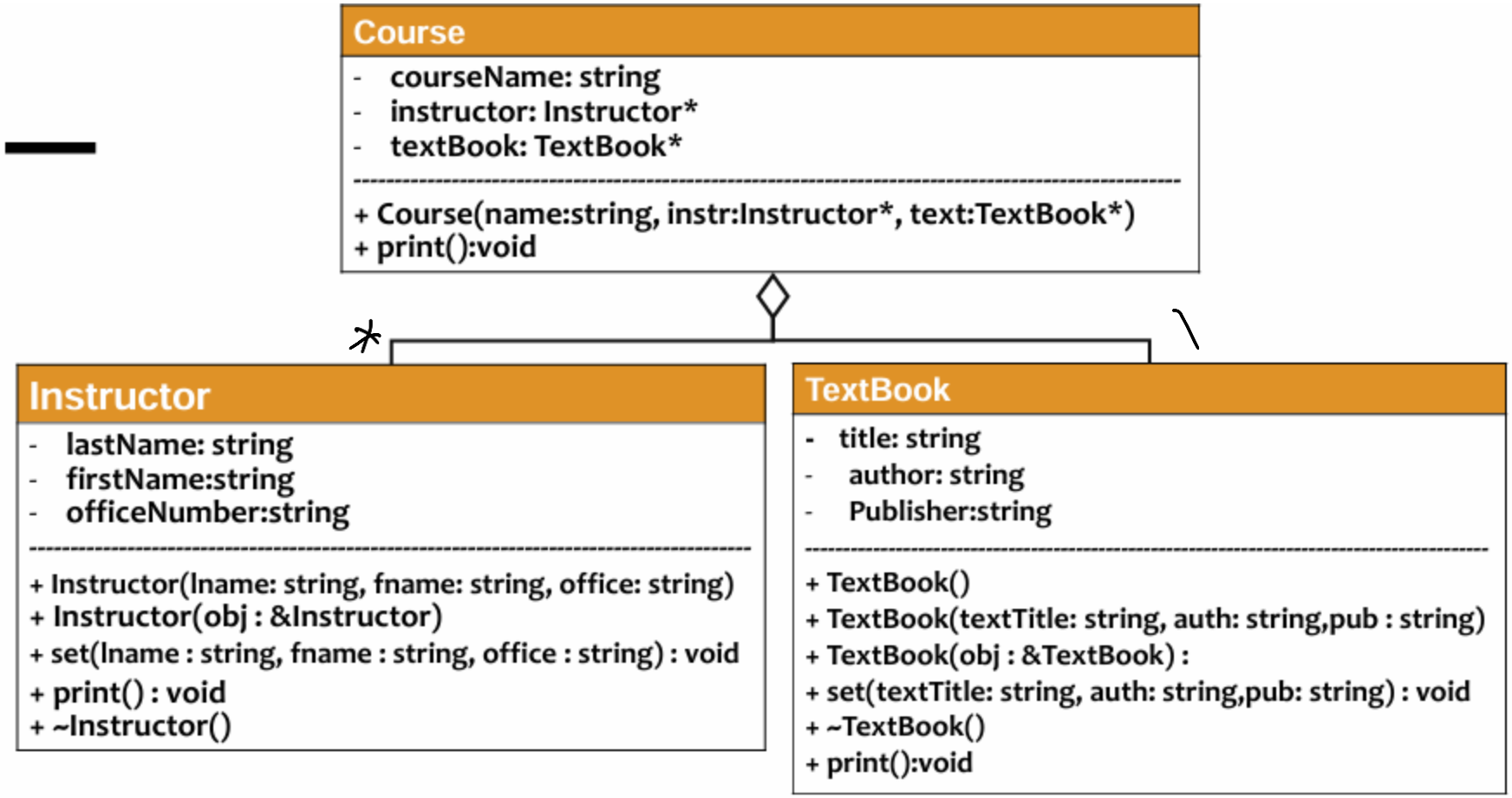
A) A class member, a class object B) A class, a member of that class

C) A class object, a member of that class D) A class object, a class

delete keyword doesn’t delete the pointer itself, it deletes what the pointer refers to.

A) True B) False


Employee
- firstName: string
- lastName: string
- birthDate: Date*
- hireDate: Date*
+ Employee (string, string, Date*, Date*)
+ print ():void
+ ~Employee ():
1
Date
- month: int
- day: int
- year: int
+ Date (int, int, int)
+ print ():void


Course
- courseName: string
- instructor: Instructor*
- textBook: TextBook*
+ Course(name:string, instr:Instructor*, text:TextBook*)
+ print():void
*
1
Instructor
- lastName: string
- firstName:string
- officeNumber:string
+ Instructor(lname: string, fname: string, office: string)
+ Instructor(obj : &Instructor)
+ set(lname : string, fname : string, office : string) : void
+ print() : void
+ ~Instructor()
TextBook
- title: string
- author: string
- Publisher:string
+ TextBook()
+ TextBook(textTitle: string, auth: string,pub : string)
+ TextBook(obj : &TextBook) :
+ set(textTitle: string, auth: string,pub: string) : void
+ ~TextBook()
+ print():void

What is wrong with the following program?.

```
int main( ){
      int x =20;
      int* p;
      p = new int;
      *p = 30;
      cout << *p << " " << x << endl;
      delete p;
      p = &x;
      delete p;
      cout << *p << " " << x << endl;
            return 0;
}
```

What is wrong with the following program?.

```
int main( ){
            const int SIZE = 10000000;
            int i;
            int A[SIZE];

            for(i=0; i<SIZE; i++)
                  A[i] = i;
            int *B;
            B = new int[SIZE];

            for(i=0; i<SIZE; i++)
                  B[i] = A[i];

            for(i=0; i<SIZE; i++)
                  cout << A[i] << " " << B[i] << endl;
                 return 0;
}
```

What is wrong with the following program?

```
void F1(int* temp){ *temp = 99; }
void main( ){
      int *p1, *p2;
      p1 = new int;
      *p1 = 50;
      p2 = p1;
      F1(p2);
      cout << *p1 << " " << *p2 << endl;
      p1 = new int;
      *p1 = 88;
      cout << *p1 << " " << *p2 << endl;
      delete p1;
      delete p2;
}
```

What is wrong with the following program?.

```
int main( ){
      int i, *p, *r, t[4]={0,1,2,3};

            p = new int[4];
            for(i=0; i<4; i++)
              p[i] = 6*(i+1);
            r = t;
            for(i=3; i>=0; i--)
              r[i] -= 1;
            r[2] = 8;
            for(i=0; i<4; i++)
              t[i] = p[i] + r[i];
            for(i=0; i<4; i++)
              cout << t[i] << " ";
            delete [] p;
           return 0;
      }
```

What is wrong with the following program?.

```
int main( ){
      int *p, *r;
      p = new int;
      r = new int;
      p = 1;
      r = p;
     return 0;
}
```